

Camera Capture Using OV7675 Camera

Objective:

To capture an image frame using the OV7675 camera module connected to the Arduino Nano 33 BLE and transmit the pixel data through the Serial Monitor.

(Note: [Install Arduino_OV767X Library](#))

Procedure:

1. Connected the OV7675 camera module to the Arduino Nano 33 BLE.
2. Installed the Arduino_OV767X library.
3. Opened the CameraCapture example:
4. **File → Examples → Arduino_OV767X → CameraCapture**
5. Uploaded the code to the board.
6. Opened the Serial Monitor.
7. Sent the character 'c' through the Serial Monitor.
8. The camera captured one frame and transmitted the image data as a hexadecimal string.

Working Principle:

The OV7675 camera captures an image and converts it into pixel values.

The image is captured in

- Resolution: **176 × 144 pixels (QCIF)**
- Format: **RGB565**
- Pixel Size: **16 bits per pixel**

Each pixel contains color information represented as hexadecimal values. These pixel values are transmitted through the Serial Monitor.

Observations:

- Camera successfully captured image data.
- Pixel values were transmitted as hexadecimal numbers.
- The captured data represents the image frame.

Output Serial Monitor X

Message (Enter to send message to 'Arduino Nano 33 BLE' on 'COM4')

OV767X Camera Capture

Camera settings:

width = 176

height = 144

bits per pixel = 16

Send the 'c' character to read a frame ...

Reading frame

E1102112211201120112C110C110411A6439E841A73987396739453125314631C61081108110811